Assignment Instruction: Global Data & AI Workforce Compensation Analytics

Project Overview

In this project, you will act as a senior workforce analytics consultant for a global tech executive search and recruitment firm. Your task is to, using provided data, deliver a clear and interpretable presentation to directionally and visually address critical organizational research questions regarding compensation, geographic arbitrage, and talent retention in the Data Science/AI labor market. Your data visualization should be supplemented by research from trusted sources on labor market trends, and every assertation in your presentation must be sourced with Full Note Chicago-Style citations (reference: https://libguides.westsoundacademy.org/chicago-citation/basic-web-page)

Core Business Questions

You organization has assigned you and a few teammates to this project in order to visually address the following stakeholder broad lines of inquiry, which you can clarify in stakeholder meetings during office hours:

- Where are salaries growing?
- How do compensation trajectories differ across career levels?
- Is there a "remote work premium" or "remote penalty"?
- What is the financial return when progressing through career tiers (Entry → Mid → Senior → Executive)?
- Research component: based on your answers above, propose a market-leading Fortune 500 publicly-traded US company that our recruiting firm should approach who has growth needs for resources related to our data set, and make a persuasive case to our internal steering committee as to what specific job titles and levels of seniority we should prioritize as we grow our pipeline of candidates to offer to our clients.

PART I: Data Architecture & Sources

Ingest, clean, standardize, and query data across historical timelines and regional boundaries:

1. Core Workforce Datasets (Public Domain)

- **Data Science & AI Job Salaries (2020–2026):** Consolidated multi-year global records capturing job titles, experience levels, employment types, USD-equivalent converted salaries, employee residences, remote work ratios, and company headquarters locations.
    - *Data Source:* Kaggle Data Science Job Salaries Dataset

PART II: Technical Reporting

Your final work products must feature at least one of the following visualizations within an interactive dashboard or reproducible report:

1. The Compensation Distribution & Outlier Dashboard

- Create a scannable visualization that isolates extreme income outliers (e.g., compensation $\geq \$500{,}000$).
- Display log-transformed distributions and interquartile range (IQR) boxplots across role domains to demonstrate the variance and mathematical skewness of high-tier tech compensation.

2. Normalized Geographic Arbitrage Map

- Construct a global map displaying median salaries by employee residence versus employer location.
- Implement a toggle or side-by-side view that contrasts localized domestic pay against cross-border remote pay for US-based employers, highlighting regions benefiting from remote capital influx.

3. Career Progression & Experience Playbook

- Produce a faceted time-series or heatmap chart showing salary progression across experience tiers (EN,MI,SE,EX) segmented by company size (Small,Medium,Large).
- Calculate and display the percentage margin gains $(\text{SE} - \text{MI})/\text{MI}$ to assist our organization's clients (inclusive of HR compensation committees responsible for workforce level-planning and pay-band design).

You will prepare a presentation for an audience of internal executives to communicate insights from the data set relevant to the stakeholder interests articulated above. Explaining in your oral presentation why you made visualization design decisions you made, and how you collaborated within your team. Address all core business questions.

PRO SKILLS HW #6 - Your presentation is due in Canvas for Pro Skills class on 10/20 at 8am (Homework #6). Clarifications from your sponsor (representing the internal executive team) will make himself available for clarification and questions during 36611 Open Office Hours on 9/22, 9/29, and 10/6 from 12pm – 1pm in Wean 3509. No clarifications will be provided outside of these sponsor information sessions.